

STD 2: Networks (Std 2), N2 Network Concepts (Y12)

Shortest Paths

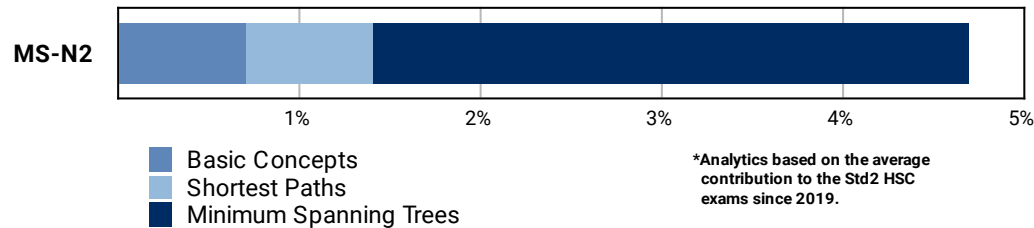
Basic Concepts

Minimum Spanning Trees

Teacher: Davidson Lai

Exam Equivalent Time: 174 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

STD2 Exam Contribution History N2 Network Concepts



OVERVIEW

- *Networks* is a new Std2 topic area and has been allocated between 9-13% of the Std2 exam over the first 3 years of the new syllabus.
- We have split *N2 Network Concepts* (4.7%) into three categories for the purposes of analysis: *Basic Concepts* (0.7%), *Shortest Paths* (0.7%) and *Minimum Spanning Trees* (3.3%).
- This analysis will look at *Shortest Paths*.

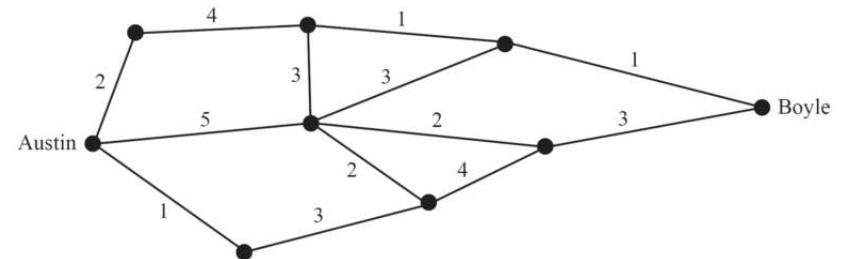
ANALYSIS -

- This topic area has typically been associated with low to mid-band style questions from our research.
- This area was tested in 2021 and 2019 in simple 1-mark questions that were part of more substantial minimum spanning tree questions.
- Implementing *Dijkstra's algorithm* to solve questions in this topic has been a focus. This solution method is provided in detail in most worked solutions, reflecting its importance in NESAs *Networks Topic Guide*.
- Questions can require students to find a "shortest distance" by providing either a network diagram or a table, both of which are covered in the database.

Questions

1. Networks, STD2 N2 2009 FUR1 2 MC

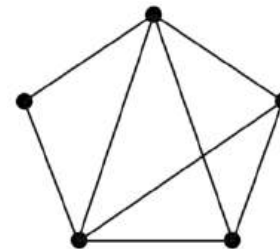
The network shows the distances, in kilometres, along roads that connect the cities of Austin and Boyle.



The shortest distance, in kilometres, from Austin to Boyle is

- A. 7
- B. 8
- C. 9
- D. 10

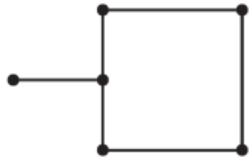
2. Networks, STD2 N2 2010 FUR1 2 MC



The number of edges in the graph above is

- A. 5
- B. 7
- C. 8
- D. 10

3. Networks, STD2 N2 2012 FUR1 1 MC



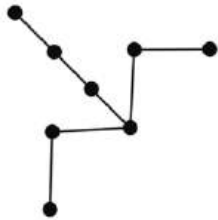
The sum of the degrees of all the vertices in the graph above is

- A. 6
- B. 9
- C. 11
- D. 12

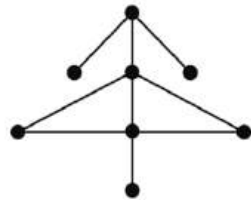
4. Networks, STD2 N2 2013 FUR1 1 MC

Which one of the following graphs is a tree?

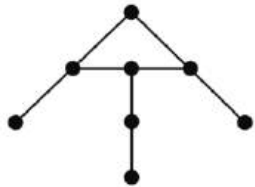
A.



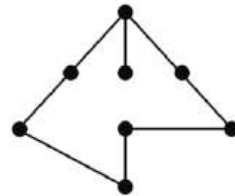
B.



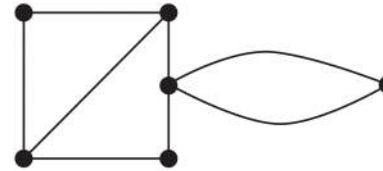
C.



D.



5. Networks, STD2 N2 2015 FUR1 1 MC



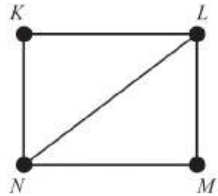
In the graph above, the number of vertices of odd degree is

- A. 1
- B. 2
- C. 3
- D. 4

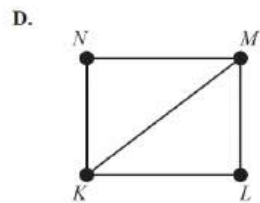
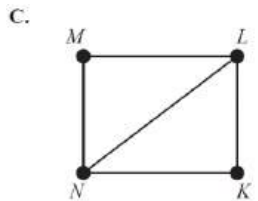
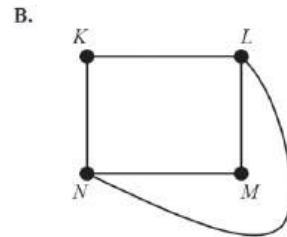
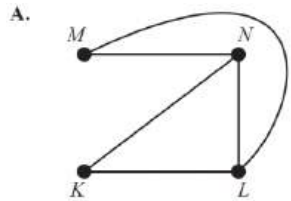
6. Networks, STD2 N2 2015 FUR1 5 MC

The graph below represents a friendship network. The vertices represent the four people in the friendship network: Kwan (K), Louise (L), Milly (M) and Narelle (N).

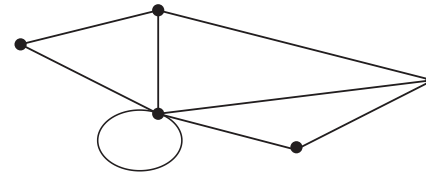
An edge represents the presence of a friendship between a pair of these people. For example, the edge connecting K and L shows that Kwan and Louise are friends.



Which one of the following graphs does not contain the same information?



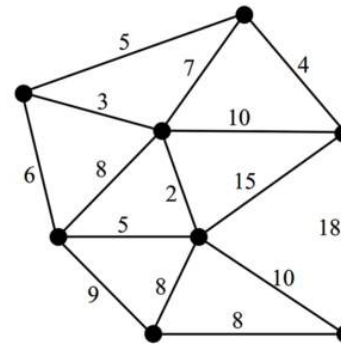
7. Networks, STD2 N2 SM-Bank 32 MC



The number of vertices with an odd degree in the network above is

- A. 2
- B. 3
- C. 4
- D. 5

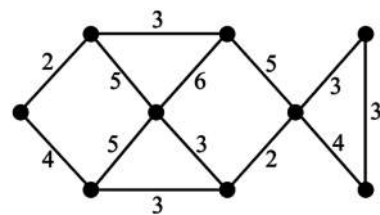
8. Networks, STD2 N2 2010 FUR1 5 MC



For the network above, the length of the minimal spanning tree is

- A. 30
- B. 31
- C. 35
- D. 39

9. Networks, STD2 N2 2013 FUR1 3 MC

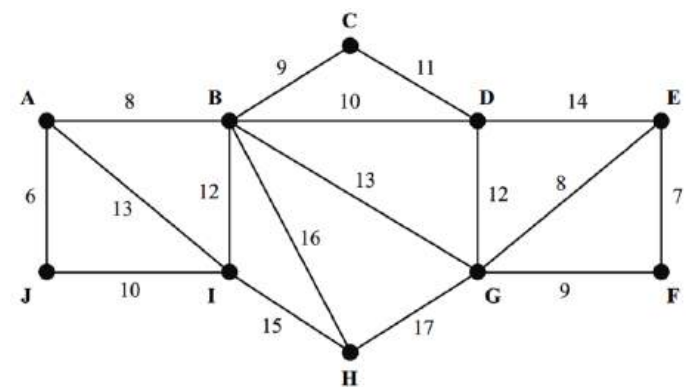


The vertices of the graph above represent nine computers in a building. The computers are to be connected with optical fibre cables, which are represented by edges. The numbers on the edges show the costs, in hundreds of dollars, of linking these computers with optical fibre cables.

Based on the same set of vertices and edges, which one of the following graphs shows the cable layout (in bold) that would link all the computers with optical fibre cables for the minimum cost?

- A.
- B.
- C.
- D.

10. Networks, STD2 N2 2014 FUR1 5 MC

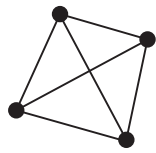


Which one of the following is the minimal spanning tree for the weighted graph shown above?

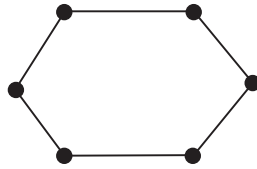
- A.
- B.
- C.
- D.

11. Networks, STD2 N2 2017 FUR1 2 MC

Two graphs, labelled Graph 1 and Graph 2, are shown below.



Graph 1



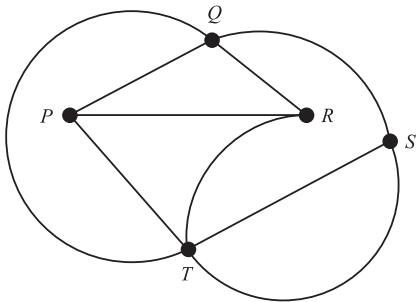
Graph 2

The sum of the degrees of the vertices of Graph 1 is

- A. two less than the sum of the degrees of the vertices of Graph 2.
- B. one less than the sum of the degrees of the vertices of Graph 2.
- C. equal to the sum of the degrees of the vertices of Graph 2.
- D. two more than the sum of the degrees of the vertices of Graph 2.

12. Networks, STD2 N2 2018 FUR1 4 MC

Consider the graph below.



Which one of the following is **not** a path for this graph?

- A. *PRQTS*
- B. *PRTSQ*
- C. *PTQSR*
- D. *PTRQS*

13. Networks, STD2 N2 SM-Bank 3 MC

A store manager is directly in charge of five department managers.

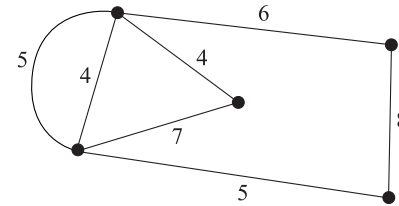
Each department manager is directly in charge of six sales people in their department.

This staffing structure could be represented graphically by

- A. a tree.
- B. a path.
- C. a cycle.
- D. a weighted graph.

14. Networks, STD2 N2 SM-Bank 26 MC

A weighted network diagram is shown below.

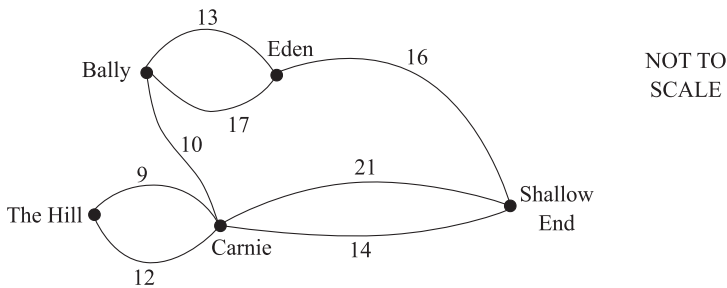


What is the weight of the minimum spanning tree?

- A. 18
- B. 19
- C. 20
- D. 22

15. Networks, STD2 N2 SM-Bank 27 MC

This diagram shows the possible paths (in km) for laying gas pipes between various locations.

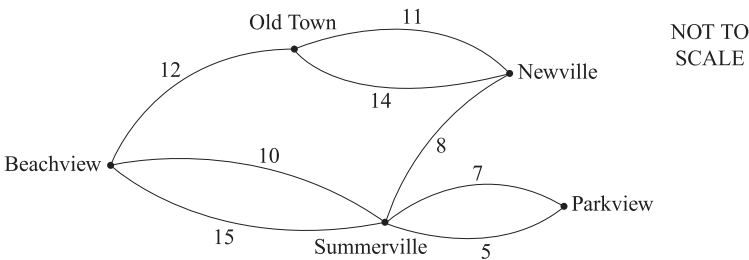


Gas is to be supplied from one location. Any one of the locations can be the source of the supply.
What is the minimum total length of length of the pipes required to provide gas to all the locations?

- A. 46 km
- B. 48 km
- C. 50 km
- D. 52 km

16. Networks, STD2 N2 SM-Bank 1 MC

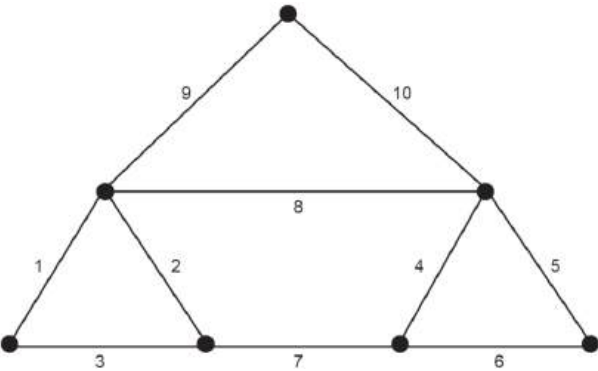
This diagram shows the possible paths (in km) for laying gas pipes between various locations.



Gas is to be supplied from one location. Any one of the locations can be the source of the supply.
What is the minimum total length of the pipes required to provide gas to all the locations?

- A. 32 km
- B. 34 km
- C. 36 km
- D. 38 km

17. Networks, STD2 N2 2006 FUR1 4 MC

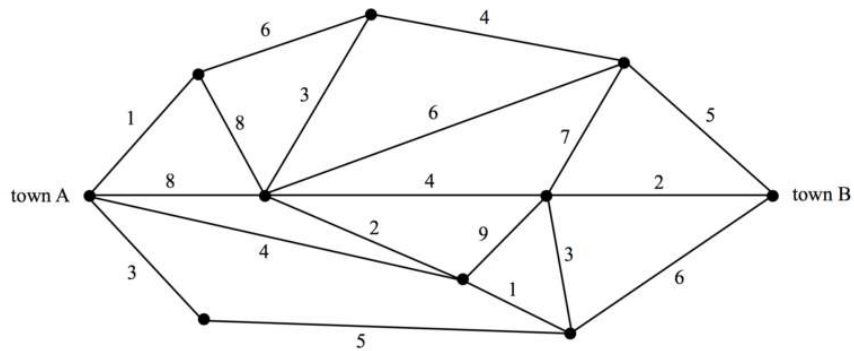


The minimal spanning tree for the network above will include the edge that has a weight of

- A. 3
- B. 6
- C. 8
- D. 9

18. Networks, STD2 N2 2007 FUR1 4 MC

The following network shows the distances, in kilometres, along a series of roads that connect town **A** to town **B**.

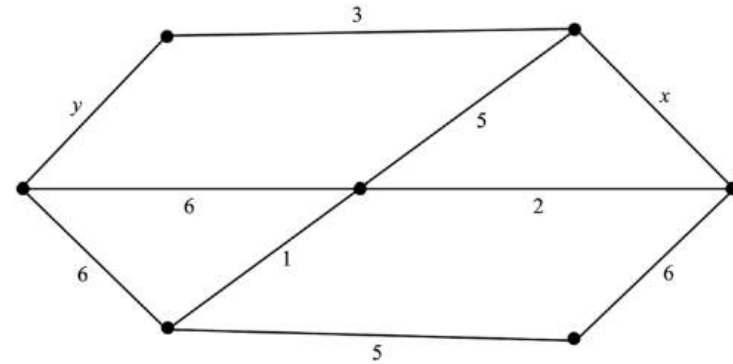


The shortest distance, in kilometres, to travel from town **A** to town **B** is

- A. 9
- B. 10
- C. 11
- D. 12

19. Networks, STD2 N2 2007 FUR1 7 MC

The minimal spanning tree for the network below includes two edges with weightings ***x*** and ***y***.

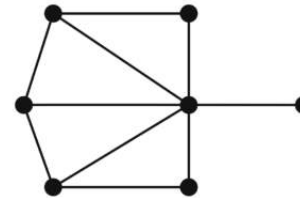


The length of the minimal spanning tree is 19.

The values of ***x*** and ***y*** could be

- A. ***x* = 1 and *y* = 7**
- B. ***x* = 2 and *y* = 5**
- C. ***x* = 3 and *y* = 5**
- D. ***x* = 4 and *y* = 5**

20. Networks, STD2 N2 2011 FUR1 1 MC

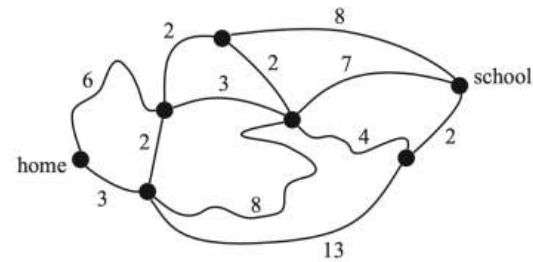


In the network shown, the number of vertices of even degree is

- A. 2
- B. 3
- C. 4
- D. 5

21. Networks, STD2 N2 2014 FUR1 3 MC

The diagram below shows the network of roads that Stephanie can use to travel between home and school.



The numbers on the roads show the time, in minutes, that it takes her to ride a bicycle along each road.

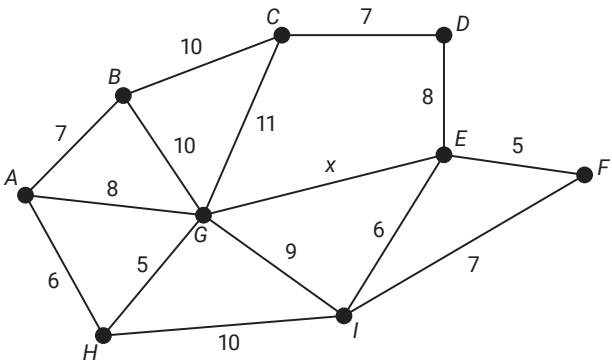
Using this network of roads, the shortest time that it will take Stephanie to ride her bicycle from home to school is

- A. 12 minutes
- B. 13 minutes
- C. 14 minutes
- D. 15 minutes

22. Networks, STD2 N2 2020 FUR1 5 MC

The network below shows the distances, in metres, between camp sites at a camping ground that has electricity.

The vertices **A** to **I** represent the camp sites.



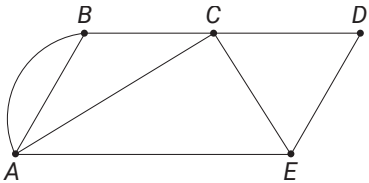
The minimum length of cable required to connect all the camp sites is 53 m.

The value of **x**, in metres, is at least

- A. 6
- B. 8
- C. 9
- D. 11

23. Networks, STD2 N2 2021 HSC 2 MC

Consider the network diagram.



What is the sum of the degrees of all the vertices in this network?

- A. 5
- B. 8
- C. 14
- D. 16

24. Networks, STD2 N2 SM-Bank 25 MC

In a town, there are four cafes **W**, **X**, **Y** and **Z**. The table shows the distances, in metres, of paved footpath connecting the cafes.

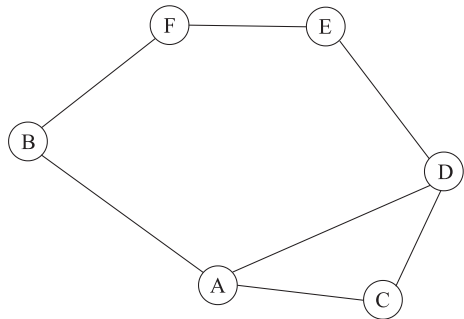
	<i>W</i>	<i>X</i>	<i>Y</i>	<i>Z</i>
<i>W</i>	—	150	—	120
<i>X</i>	150	—	140	70
<i>Y</i>	—	140	—	100
<i>Z</i>	120	70	100	—

A coffee supplier needs to visit each cafe.

What is the shortest distance she needs to walk along the paved footpath if she starts at cafe **W**?

- A. 260 m
- B. 320 m
- C. 330 m
- D. 360 m

25. Networks, STD2 N2 SM-Bank 7 MC

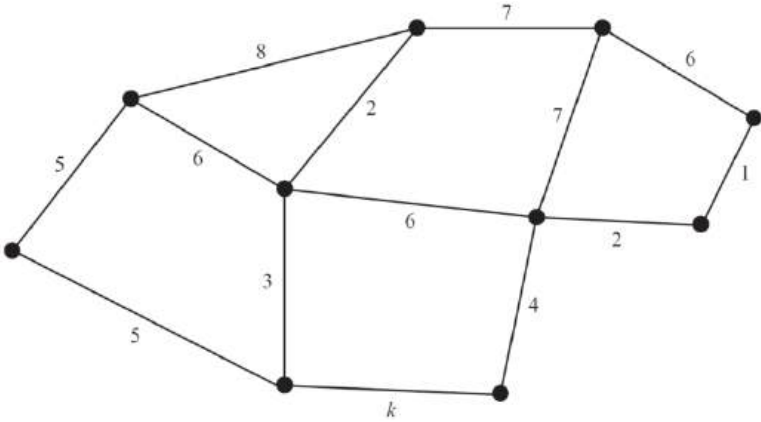


How many spanning trees are possible for this network?

- A. 3
- B. 8
- C. 14
- D. 30

26. Networks, STD2 N2 2016 FUR1 4 MC

The minimum spanning tree for the network below includes the edge with weight labelled ***k***.



The total weight of all edges for the minimum spanning tree is 33.

The value of ***k*** is

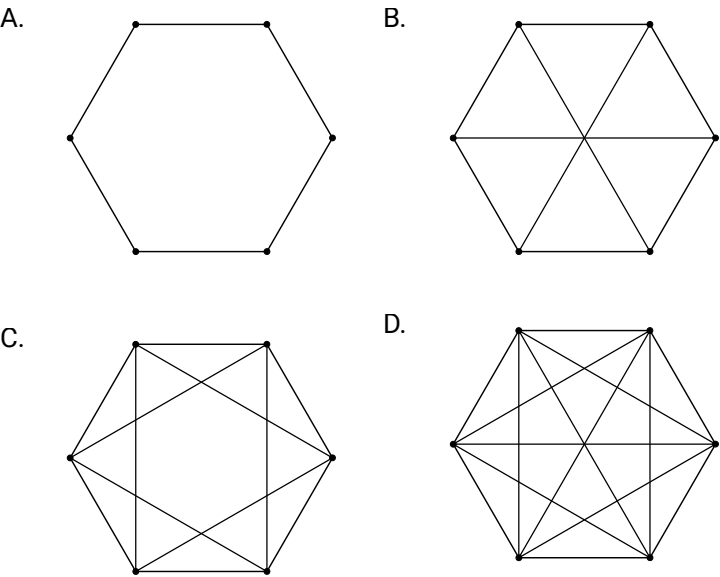
- A. 2
- B. 3
- C. 4
- D. 5

27. Networks, STD2 N2 2020 HSC 9 MC

Team **A** and Team **B** have entered a chess competition.

Team **A** and **B** have three members each. Each member of Team **A** must play each member of Team **B** once.

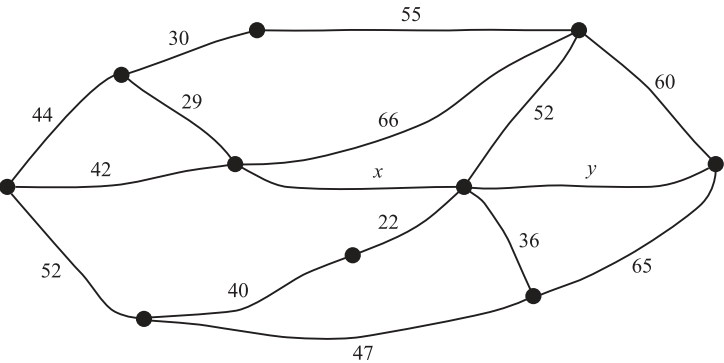
Which of the following network diagrams could represent the chess games to be played?



28. Networks, STD2 N2 2021 FUR1 8 MC

A network of roads connecting towns in an alpine region is shown below.

The distances between neighbouring towns, represented by vertices, are given in kilometres.



The region receives a large snowfall, leaving all roads between the towns closed to traffic.

To ensure each town is accessible by car from every other town, some roads will be cleared.

The minimal total length of road, in kilometres, that needs to be cleared is

- A. 361 if $x = 50$ and $y = 55$
- B. 361 if $x = 50$ and $y = 60$
- C. 366 if $x = 55$ and $y = 55$
- D. 371 if $x = 55$ and $y = 65$

29. Networks, STD2 N2 2009 FUR1 8 MC

An undirected connected graph has five vertices.

Three of these vertices are of even degree and two of these vertices are of odd degree.

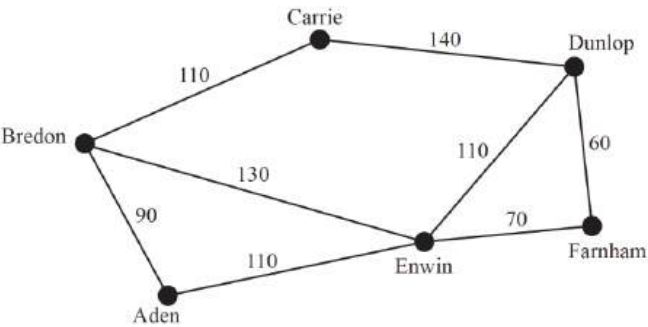
One extra edge is added. It joins two of the existing vertices.

In the resulting graph, it is **not** possible to have five vertices that are

- A. all of even degree.
- B. all of equal degree.
- C. one of even degree and four of odd degree.
- D. four of even degree and one of odd degree.

30. Networks, STD2 N2 2011 FUR2 1

Aden, Bredon, Carrie, Dunlop, Enwin and Farnham are six towns.
The network shows the road connections and distances between these towns in kilometres.



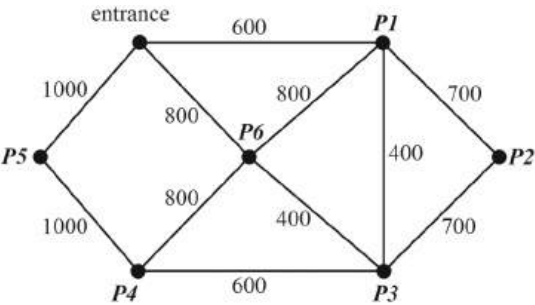
- a. In kilometres, what is the shortest distance between Farnham and Carrie? (1 mark)
- b. How many different ways are there to travel from Farnham to Carrie without passing through any town more than once? (1 mark)

An engineer plans to inspect all of the roads in this network.
He will start at Dunlop and inspect each road only once.

- c. At which town will the inspection finish? (1 mark)

31. Networks, STD2 N2 2013 FUR2 1

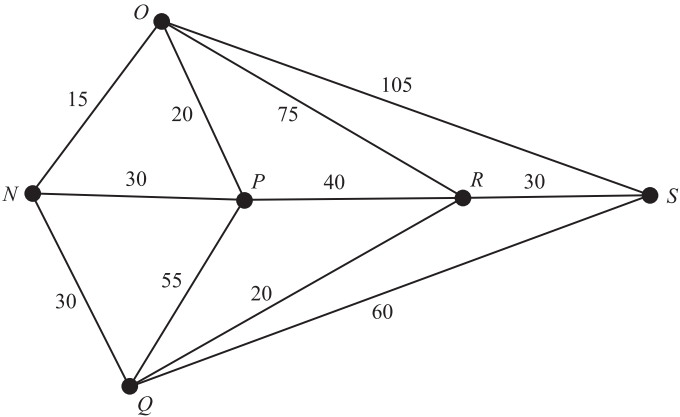
The vertices in the network diagram below show the entrance to a wildlife park and six picnic areas in the park: **P1**, **P2**, **P3**, **P4**, **P5** and **P6**.
The numbers on the edges represent the lengths, in metres, of the roads joining these locations.



- a. In this graph, what is the degree of the vertex at the entrance to the wildlife park? (1 mark)
- b. What is the shortest distance, in metres, from the entrance to picnic area **P3**? (1 mark)

32. Networks, STD2 N2 2017 FUR2 1

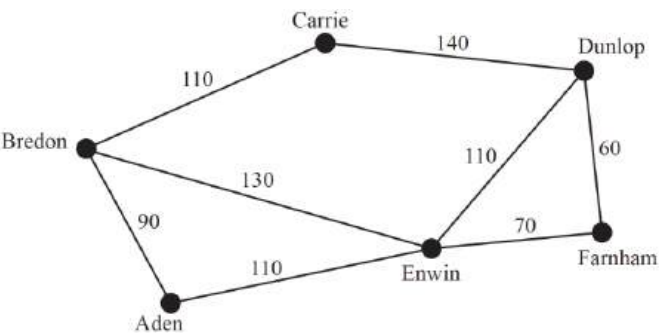
Bus routes connect six towns.
The towns are Northend (**N**), Opera (**O**), Palmer (**P**), Quigley (**Q**), Rosebush (**R**) and Seatown (**S**).
The graph below gives the cost, in dollars, of bus travel along these routes.
Bai lives in Northend (**N**) and he will travel by bus to take a holiday in Seatown (**S**).



- a. Bai considers travelling by bus along the route Northend (**N**) – Opera (**O**) – Seatown (**S**).
How much would Bai have to pay? (1 mark)
- b. If Bai takes the cheapest route from Northend (**N**) to Seatown (**S**), which other town(s) will he pass through? (1 mark)

33. Networks, STD2 N2 2011 FUR2 1

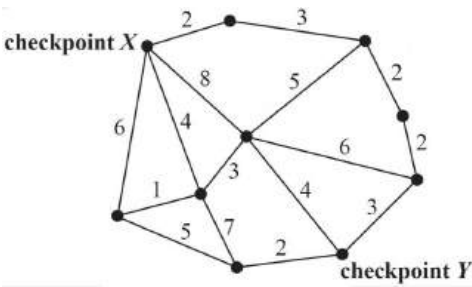
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The network shows the road connections and distances between these towns in kilometres.



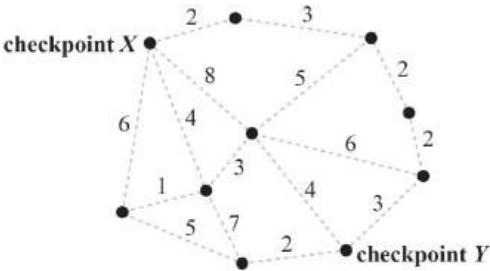
- a. In kilometres, what is the shortest distance between Farnham and Carrie? (1 mark)
- b. How many different ways are there to travel from Farnham to Carrie without passing through any town more than once? (1 mark)

34. Networks, STD2 N2 2010 FUR2 2

The diagram below shows a network of tracks (represented by edges) between checkpoints (represented by vertices) in a short-distance running course. The numbers on the edges indicate the time, in minutes, a team would take to run along each track.

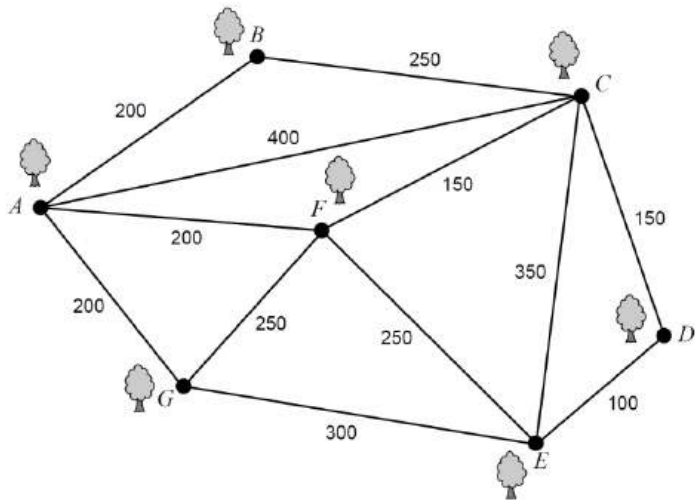


- A challenge requires teams to run from checkpoint **X** to checkpoint **Y** using these tracks.
- What would be the shortest possible time for a team to run from checkpoint **X** to checkpoint **Y**? Mark the shortest route on the diagram below. (2 marks)



35. Networks, STD2 N2 SM-Bank 13

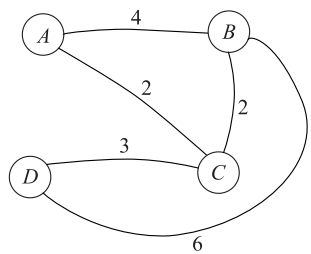
An estate has large open parklands that contain seven large trees.
The trees are denoted as vertices A to G on the network diagram below.
Walking paths link the trees as shown.
The numbers on the edges represent the lengths of the paths in metres.



Jamie is standing at A and Michelle is standing at D.
Write down the shortest route that Jamie can take and the distance travelled to meet Michelle at D.
(1 mark)

36. Networks, STD2 N2 SM-Bank 20

A table is constructed to represent the network diagram below.



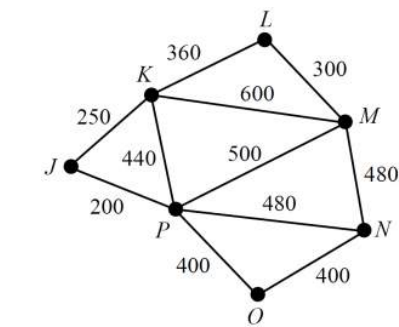
Complete the table. (2 marks)

	A	B	C	D
A	0	4		-
B	4	0		
C			0	3
D		6		0

37. Networks, STD2 N2 2015 FUR2 1

A factory requires seven computer servers to communicate with each other through a connected network of cables.

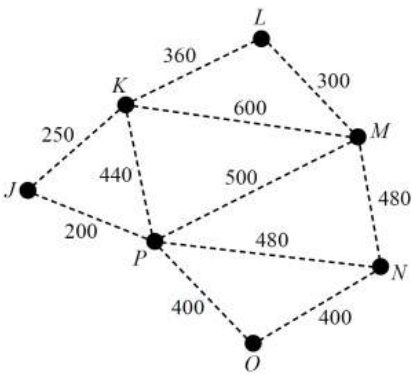
The servers, **J**, **K**, **L**, **M**, **N**, **O** and **P**, are shown as vertices on the graph below.



The edges on the graph represent the cables that could connect adjacent computer servers.

The numbers on the edges show the cost, in dollars, of installing each cable.

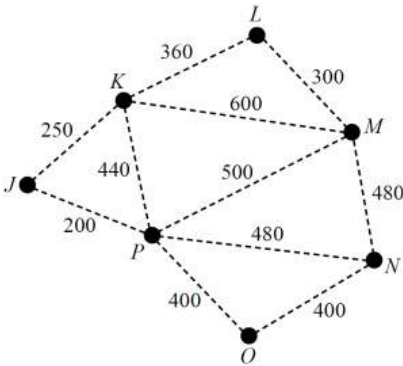
- a. What is the cost, in dollars, of installing the cable between server **L** and server **M**? (1 mark)
- b. What is the cheapest cost, in dollars, of installing cables between server **K** and server **N**? (1 mark)
- c. An inspector checks the cables by walking along the length of each cable in one continuous path. To avoid walking along any of the cables more than once, at which vertex should the inspector start and where would the inspector finish? (1 mark)
- d. The computer servers will be able to communicate with all the other servers as long as each server is connected by cable to at least one other server.
 - i. The cheapest installation that will join the seven computer servers by cable in a connected network follows a minimum spanning tree. Draw the minimum spanning tree on the plan below. (1 mark)



- ii. The factory's manager has decided that only six connected computer servers will be needed, rather than seven.

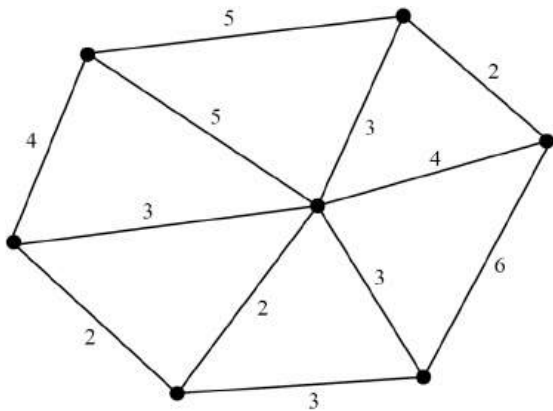
How much would be saved in installation costs if the factory removed computer server **P** from its minimum spanning tree network?

A copy of the graph above is provided below to assist with your working. (1 mark)

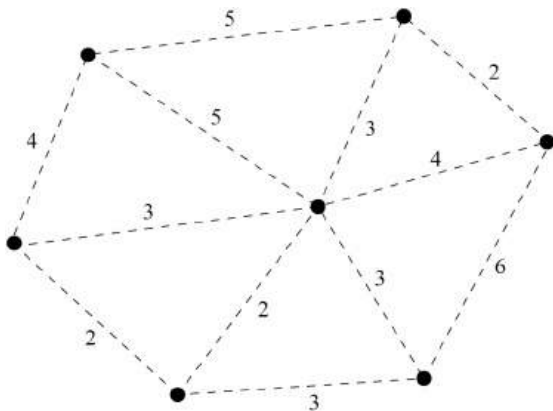


38. Networks, STD2 N2 2008 FUR2 1

James, Dante, Tahlia and Chanel are four children playing a game.
In this children’s game, seven posts are placed in the ground.
The network below shows distances, in metres, between the seven posts.
The aim of the game is to connect the posts with ribbon using the shortest length of ribbon.
This will be a minimal spanning tree.



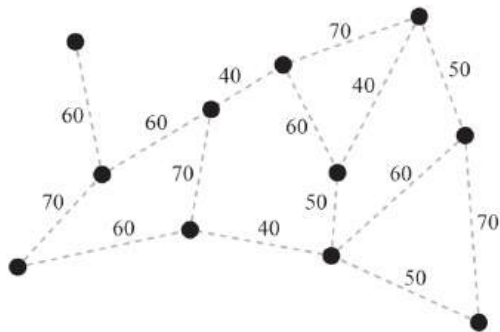
a. Draw in a minimal spanning tree for this network on the diagram below. (1 mark)



- b. Determine the length, in metres, of this minimal spanning tree. (1 mark)
- c. How many different minimal spanning trees can be drawn for this network? (1 mark)

39. Networks, STD2 N2 2011 FUR2 2

At the Farnham showgrounds, eleven locations require access to water. These locations are represented by vertices on the network diagram shown below. The dashed lines on the network diagram represent possible water pipe connections between adjacent locations. The numbers on the dashed lines show the minimum length of pipe required to connect these locations in metres.



- All locations are to be connected using the smallest total length of water pipe possible.
- a. On the diagram, show where these water pipes will be placed. (1 mark)
- b. Calculate the total length, in metres, of water pipe that is required. (1 mark)

40. Networks, STD2 N2 SM-Bank 28

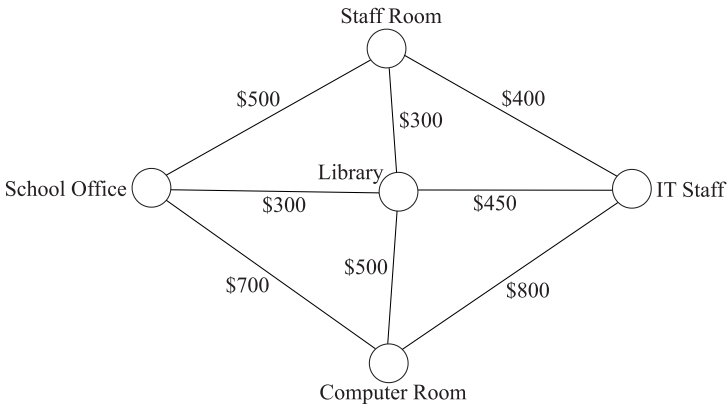
In central Queensland, there are four petrol stations **A**, **B**, **C** and **D**. The table shows the length, in kilometres, of roads connecting these petrol stations.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>A</i>	—	170	—	150
<i>B</i>	170	—	160	90
<i>C</i>	—	160	—	120
<i>D</i>	150	90	120	—

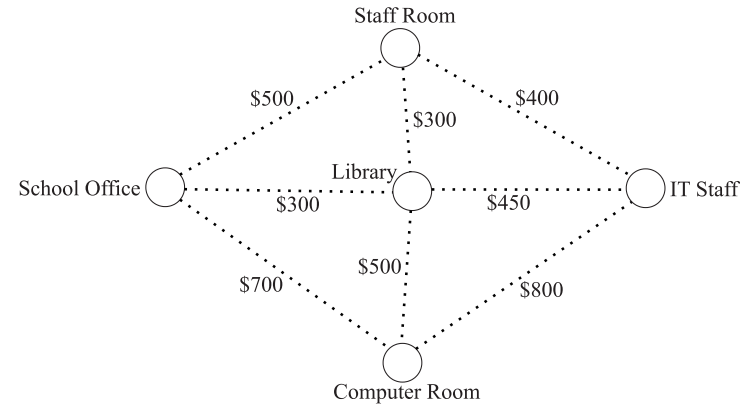
- a. Construct a network diagram to represent the information in the table. (2 marks)
- b. A petrol tanker needs to refill each station. It starts at Station **A** and visits each station.
Calculate the shortest distance that can be travelled by the petrol tanker. In your answer, include the order the petrol stations are refilled. (2 marks)

41. Networks, STD2 N2 SM-Bank 2

A school is designing a computer network between five key areas within the school.
The cost of connecting the rooms is shown in the diagram below.



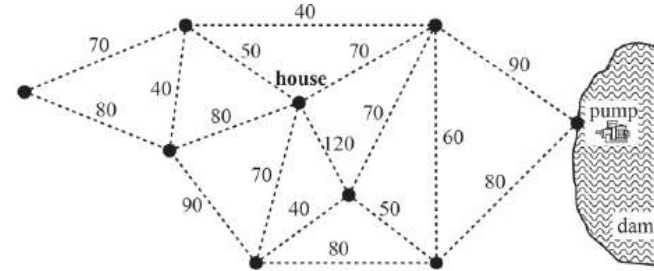
i. Complete the spanning tree below that creates the school's network at a minimum cost. (1 mark)



ii. What is the minimum cost of the network? (1 mark)

42. Networks, STD2 N2 SM-Bank 14

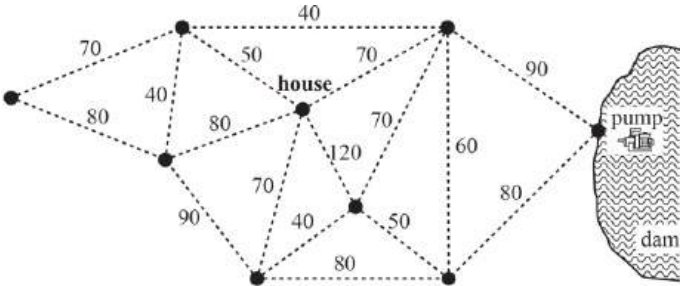
Water will be pumped from a dam to eight locations on a farm.
The pump and the eight locations (including the house) are shown as vertices in the network diagram below.
The numbers on the edges joining the vertices give the shortest distances, in metres, between locations.



i. How many vertices on the network diagram have an odd degree? (1 mark)

The total length of all edges in the network is 1180 metres.
The total length of pipe that supplies water from the pump to the eight locations on the farm is a minimum.
This minimum length of pipe is laid along some of the edges in the network.

ii. On the diagram below, **draw** the minimum length of pipe that is needed to supply water to all locations on the farm. (2 marks)



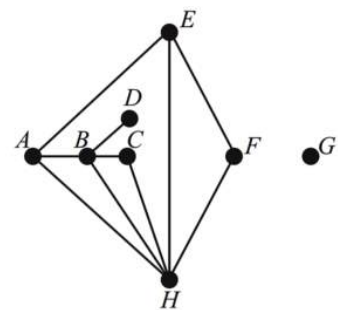
iii. What is the mathematical term that is used to describe this minimum length of pipe? (1 mark)

43. Networks, STD2 N2 SM-Bank 37

The map of Australia shows the six states, the Northern Territory and the Australian Capital Territory (ACT).



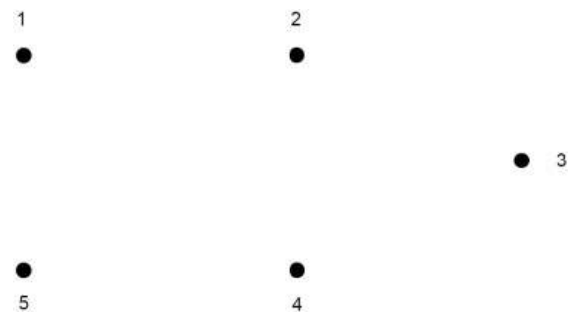
In the network diagram below, each of the vertices **A** to **H** represents one of the states or territories shown on the map of Australia. The edges represent a border shared between two states or between a state and a territory.



- i. In the network diagram, what is the order of the vertex that represents the Australian Capital Territory (ACT)? (1 mark)
- ii. In the network diagram, Queensland is represented by which letter? Explain why. (2 marks)

44. Networks, STD2 N2 2007 FUR2 1

A new housing estate is being developed.
There are five houses under construction in one location.
These houses are numbered as points 1 to 5 below.

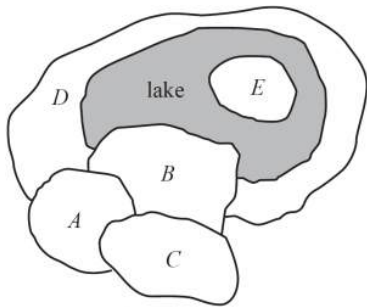


The builders require the five houses to be connected by electrical cables to enable the workers to have a supply of power on each site.

- a. What is the **minimum** number of edges needed to connect the five houses? (1 mark)
- b. On the diagram above, **draw** a connected graph with this number of edges. (1 mark)

45. Networks, STD2 N2 2009 FUR2 1

The city of Robville is divided into five suburbs labelled as **A** to **E** on the map below.
A lake which is situated in the city is shaded on the map.



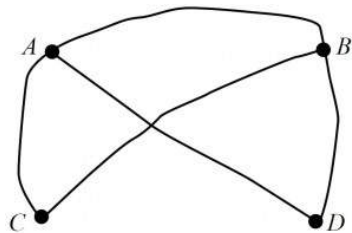
A table is constructed to represent the number of land borders between suburbs.
If there is no land border between two suburbs, the table records a '0'. If there is a single land border between two suburbs, the table records a '1', and if there are two separate land borders between the same two suburbs, the table records a '2'.

	A	B	C	D	E
A	0	1	1	1	0
B	1	0	1	2	0
C	1	1	0	0	0
D	1	2	0	0	0
E	0	0	0	0	0

a. Explain why all values in the final row and final column of the table are zero. (1 mark)

In the network diagram below, vertices represent suburbs and edges represent land borders between suburbs.

The diagram has been started but is not finished.

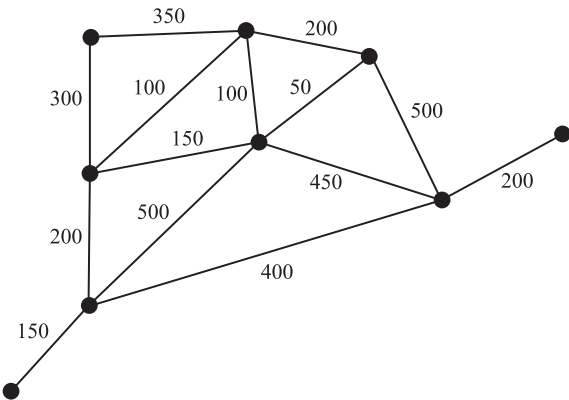


b. The network diagram is missing one edge and one vertex.
On the diagram

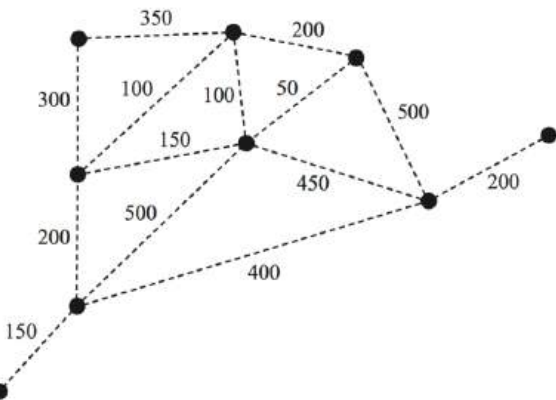
- i. draw the missing edge (1 mark)
- ii. draw and label the missing vertex. (1 mark)

46. Networks, STD2 N2 2017 FUR2 3

While on holiday, four friends visit a theme park where there are nine rides.
On the graph below, the positions of the rides are indicated by the vertices.
The numbers on the edges represent the distances, in metres, between rides.



- a. Electrical cables are required to power the rides.
These cables will form a connected graph.
The shortest total length of cable will be used.
- i. Give a mathematical term to describe a graph that represents these cables. (1 mark)
 - ii. Draw in the graph that represents these cables on the diagram below. (1 mark)



47. Networks, STD1 N1 2019 HSC 17

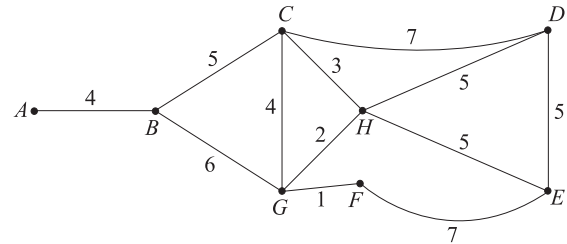
A regional airline operates flights in Queensland. Flight times between connected towns are shown in the table.

	Cairns	Kowanyama	Mt Isa	Pormpuraaw	Townsville
Cairns	–	1 h 50 min	2 h 5 min	–	55 min
Kowanyama	1 h 50 min	–	–	20 min	–
Mt Isa	2 h 5 min	–	–	–	1 h 40 min
Pormpuraaw	–	20 min	–	–	–
Townsville	55 min	–	1 h 40 min	–	–

Draw a network diagram to show how the towns are connected, with weights on the edges showing the flight times. (2 marks)

48. Networks, STD2 N2 2019 HSC 30

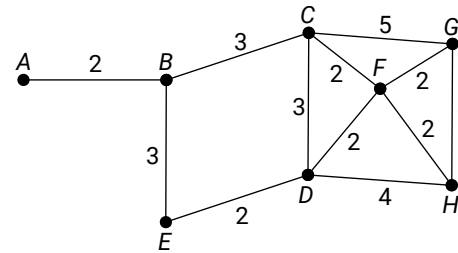
The network diagram shows the tracks connecting 8 picnic sites in a nature park. The vertices **A** to **H** represents the picnic sites. The weights on the edges represent the distance along the tracks between the picnic sites, in kilometres.



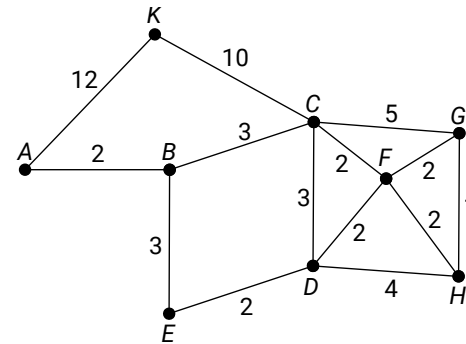
- a. Each picnic site needs to provide drinking water. The main water source is at site **A**. Draw a minimum spanning tree and calculate the minimum length of water pipes required to supply water to all the sites if the water pipes can only be laid along the tracks. (2 marks)
- b. One day, the track between **C** and **H** is closed. State the vertices that identify the shortest path from **C** to **E** that avoids the closed track. (1 mark)

49. Networks, STD2 N2 2020 HSC 18

The diagram represents a network with weighted edges.



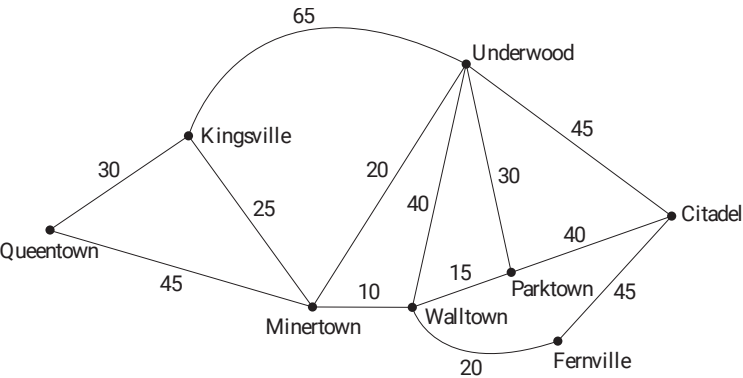
- a. Draw a minimum spanning tree for this network and determine its length. (3 marks)
- b. The network is revised by adding another vertex, **K**. Edges **AK** and **CK** have weights of 12 and 10 respectively, as shown.



What is the length of the minimum spanning tree for this revised network? (1 mark)

50. Networks, STD2 N2 2021 HSC 23

The network diagram shows the travel times in minutes along roads connecting a number of different towns.



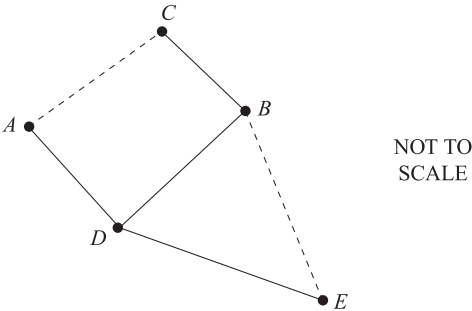
- a. Draw a minimum spanning tree for this network and determine its length. (3 marks)
- b. How long does it take to travel from Queentown to Underwood using the fastest route? (1 mark)

51. Networks, STD2 N2 SM-Bank 30

An amusement park has five toilet areas, **A**, **B**, **C**, **D** and **E**, which are connected by pathways. The table shows the length of some of the pathways, in metres.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
<i>A</i>	–	–	?	450	–
<i>B</i>	–	–	200	300	?
<i>C</i>	?	200	–	–	–
<i>D</i>	450	300	–	–	500
<i>E</i>	–	?	–	500	–

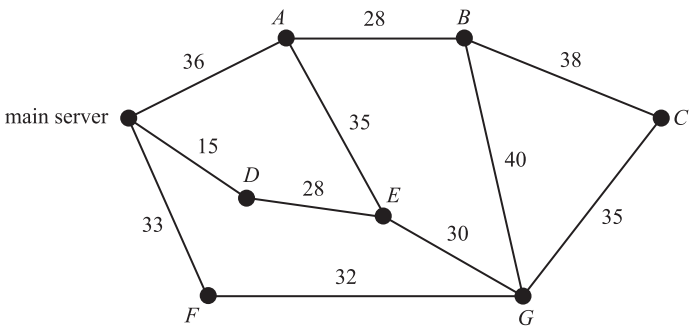
The following network diagram is drawn to represent this information and a correct minimum spanning tree is shown by the solid lines.



Complete the network diagram including a possible value for each of the two edges **AC** and **BE**, and justify why **AC** and **BE** were not included as part of the minimum spanning tree. (3 marks)

52. Networks, STD2 N2 SM-Bank 34

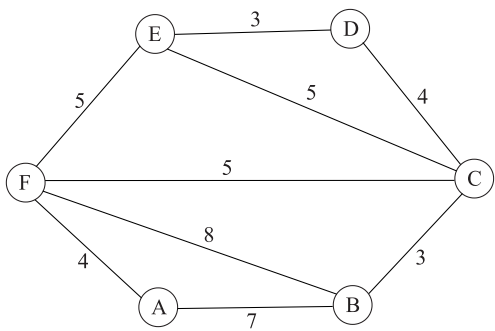
The following diagram shows the distances, in metres, along a series of cables connecting a main server to seven points, **A** to **G**, in a computer network.



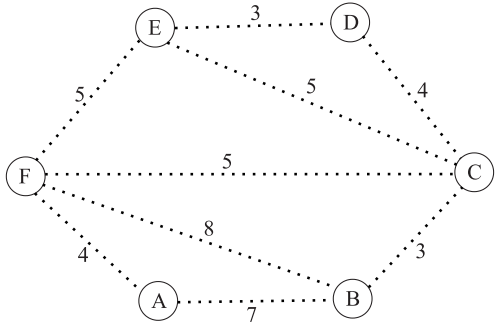
Calculate the minimum length of cable, in metres, required to ensure that each of the seven points is connected to the main server directly or via another point. **(2 marks)**

53. Networks, STD2 N2 SM-Bank 3

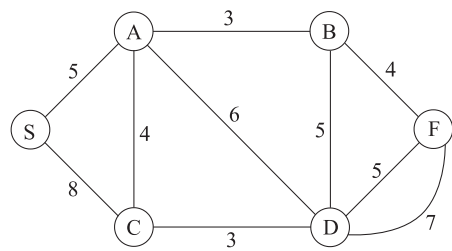
The diagram below is a connected network.



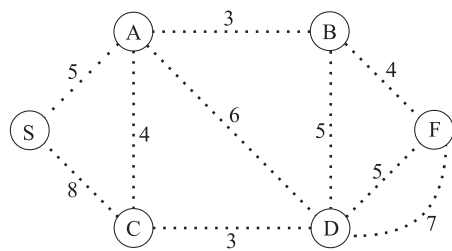
Complete the diagram below to show the minimal spanning tree of this network. **(2 marks)**



54. Networks, STD2 N2 SM-Bank 4

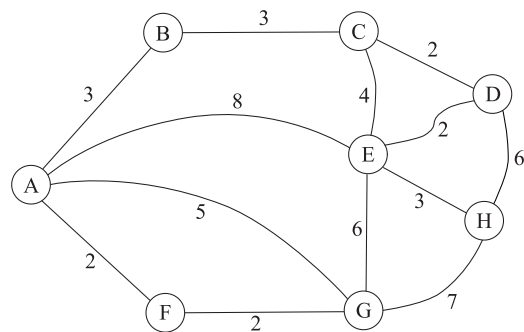


Complete the minimal spanning tree of the network above on the diagram below. (2 marks)



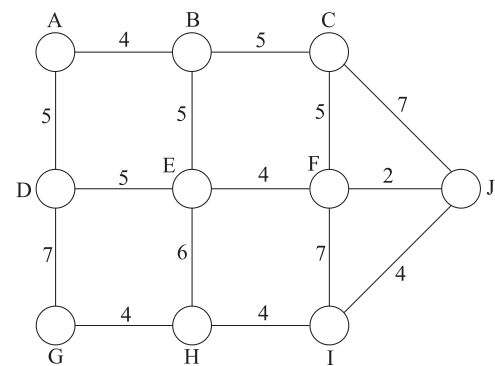
55. Networks, STD2 N2 SM-Bank 5

A network of roads is pictured below, with the distances of each road represented, in kilometres, on each edge.



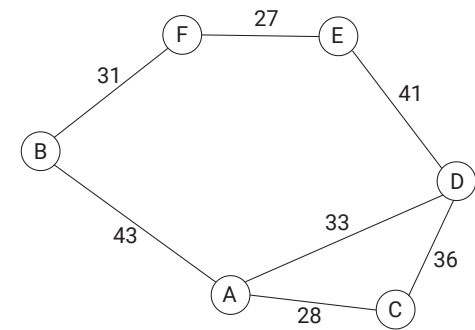
A driver wants to travel from A to H in the shortest distance possible.
Describe the possible paths she can take, and the total distance she must travel. (2 marks)

56. Networks, STD2 N2 SM-Bank 6



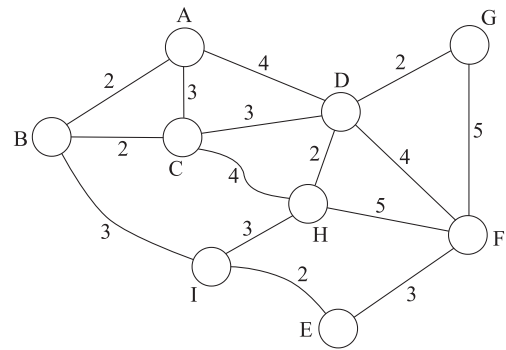
Describe the shortest path between A and J in the network above and its weight. (2 marks)

57. Networks, STD2 N2 SM-Bank 8



Highlight the minimal spanning tree of this network on the diagram above, or in a separate diagram. (2 marks)

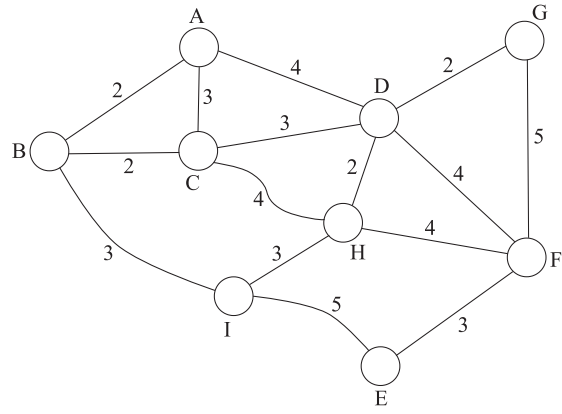
58. Networks, STD2 N2 SM-Bank 9



In a separate diagram or on the diagram above, show the minimum spanning tree . (2 marks)

59. Networks, STD2 N2 SM-Bank 10

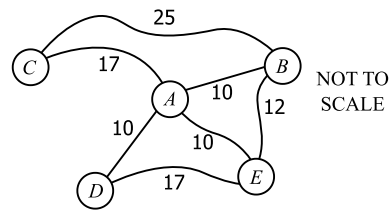
Consider the network pictured below.



Find the length of the shortest path from A to E. (2 marks)

60. Networks, STD2 N2 SM-Bank 11

A network of roads between towns shows the travelling times in minutes between towns that are directly connected.



Complete the shaded cells in the following table so that it represents the information in this network.

(2 marks)

	A	B	C	D	E
A	0	10		10	10
B		0			
C	17		0	—	
D		—	—	0	17
E	10				0

61. Networks, STD2 N2 SM-Bank 12

The following table shows the travelling time, in minutes, between towns which are directly connected by roads.

A dash indicates that towns are not directly connected.

	A	B	C	D	E
A	0	50	18	10	—
B	50	0	28	32	20
C	18	28	0	—	50
D	10	32	—	0	80
E	—	20	50	80	0

i. Draw a network diagram showing the information in this table. (2 marks)

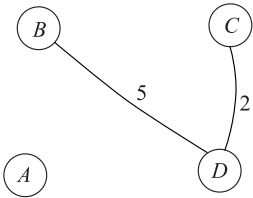
ii. What is the shortest travelling time between A and E? (1 mark)

62. Networks, STD2 N2 SM-Bank 21

The table below represents a network.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>A</i>	-	0	7	4
<i>B</i>	0	-	3	5
<i>C</i>	7	3	-	2
<i>D</i>	4	5	2	-

Complete the network diagram below to accurately reflect the information in the above table. (2 marks)

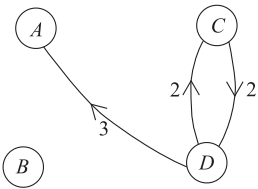


63. Networks, STD2 N2 SM-Bank 22

The table below represents a directed network.

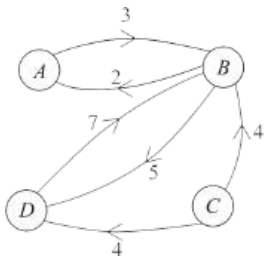
		To			
		<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
From	<i>A</i>	0	4	-	4
	<i>B</i>	-	0	-	3
	<i>C</i>	-	6	0	2
	<i>D</i>	3	5	2	0

Complete the network diagram below to accurately reflect the network described in the above table. (2 marks)



64. Networks, STD2 N2 SM-Bank 23

A directed network diagram is pictured below.



The information in the network diagram is used to complete the network table below, with a "0" used to signify that no connection exists. Complete the table. (2 marks)

		To			
		A	B	C	D
From	A	-	3		0
	B	2	-		
	C			-	4
	D				-

Worked Solutions

1. Networks, STD2 N2 2009 FUR1 2 MC

The shortest distance
= 2 + 4 + 1 + 1
= 8
⇒ B

2. Networks, STD2 N2 2010 FUR1 2 MC

⇒ C

3. Networks, STD2 N2 2012 FUR1 1 MC

Total Degrees
= 1 + 3 + 2 + 2 + 2 + 2
= 12
⇒ D

4. Networks, STD2 N2 2013 FUR1 1 MC

A tree cannot contain a cycle.
⇒ A

5. Networks, STD2 N2 2015 FUR1 1 MC

⇒ B

6. Networks, STD2 N2 2015 FUR1 5 MC

Option D has Kwan and Milly as friends which is not correct.
⇒ D

7. Networks, STD2 N2 SM-Bank 32 MC

⇒ A
(Note a loop creates 2 extra degrees to a vertex.)

8. Networks, STD2 N2 2010 FUR1 5 MC

Using Kruskal's algorithm:

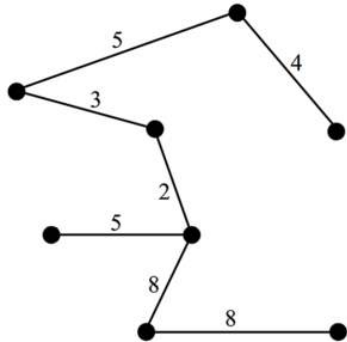
Edge 1: 2 (least weight)

Edge 2: 3

Edge 3: 4

Edges 4-5: 5

Edges 6-7: 8 (unused edges with weights <8 create circuits and are ignored)



$\therefore$ Minimal spanning tree

$$= 2 + 3 + 4 + 5 + 5 + 8 + 8$$

$$= 35$$

$\Rightarrow C$

9. Networks, STD2 N2 2013 FUR1 3 MC

Using Prim's algorithm

Starting at far left vertex,

1st edge: 2

2nd edge: 3

3rd edge: 4

4th edge: 3 etc...

$\Rightarrow A$

10. Networks, STD2 N2 2014 FUR1 5 MC

Using Prim's algorithm:

Starting at vertex A ,

1st edge: $A \rightarrow J$ (6)

2nd edge: $A \rightarrow B$ (8)

3rd edge: $B \rightarrow C$ (9)

4th edge: $J \rightarrow I$ (10)

3rd edge: $B \rightarrow D$ (10) etc...

$\Rightarrow A$

11. Networks, STD2 N2 2017 FUR1 2 MC

Graph 1

$$\sum \text{degrees} = 3 + 3 + 3 + 3 = 12$$

Graph 2

$$\sum \text{degrees} = 2 + 2 + 2 + 2 + 2 + 2 = 12$$

$\Rightarrow C$

12. Networks, STD2 N2 2018 FUR1 4 MC

By trial and error:

Consider option C ,

$PTQSR$ is not a path because S to R must go through another vertex.

$\Rightarrow C$

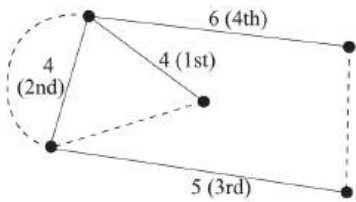
13. Networks, STD2 N2 SM-Bank 3 MC

$\Rightarrow A$

14. Networks, STD2 N2 SM-Bank 26 MC

One Strategy - Using Kruskal's Algorithm:

There are 5 vertices, so we need 4 edges.



$$\therefore \text{Weight} = 4 + 4 + 5 + 6 \\ = 19$$

$\Rightarrow B$

15. Networks, STD2 N2 SM-Bank 27 MC

Using Kruskal's Theorem:

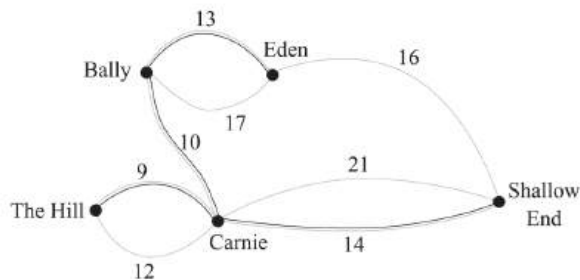
$\Rightarrow$ 5 vertices - 4 edges needed

Edge 1: The Hill $\rightarrow$ Carnie (9)

Edge 2: Carnie $\rightarrow$ Bally (10)

Edge 3: Bally $\rightarrow$ Eden (13)

Edge 4: Carnie $\rightarrow$ Shallow End (14)



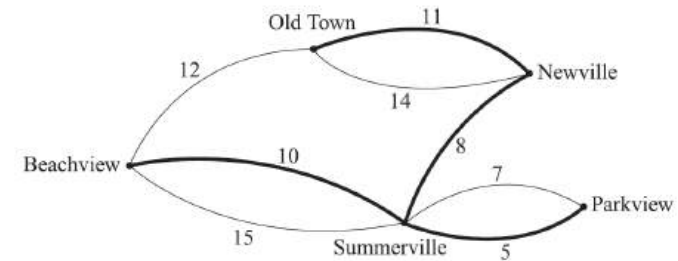
NOT TO SCALE

$$\therefore \text{Maximum length} = 9 + 10 + 13 + 14 \\ = 46 \text{ km}$$

$\Rightarrow A$

16. Networks, STD2 N2 SM-Bank 1 MC

Using Kruskal's Algorithm:



NOT TO SCALE

1st edge: Parkview - Summerville (5)

2nd edge: Summerville - Newville (8)

3rd edge: Beachview - Summerville (10)

4th edge: Old Town - Newville (11)

$\therefore$ Minimum length of pipes.

$$= 5 + 8 + 10 + 11 \\ = 34$$

$\Rightarrow B$

17. Networks, STD2 N2 2006 FUR1 4 MC

Using Kruskal's Algorithm:

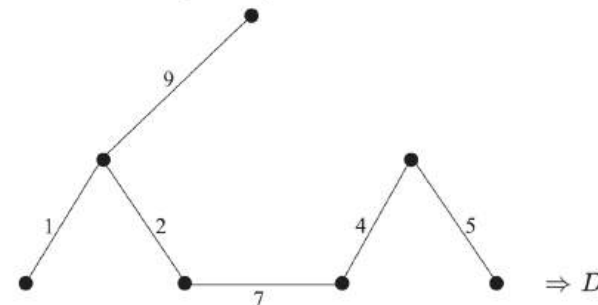
Edge 1: 1 (minimum weight)

Edge 2: 2

Edge 3: 4 (edge weight 3 creates a circuit and is ignored)

Edge 4: 5 etc...

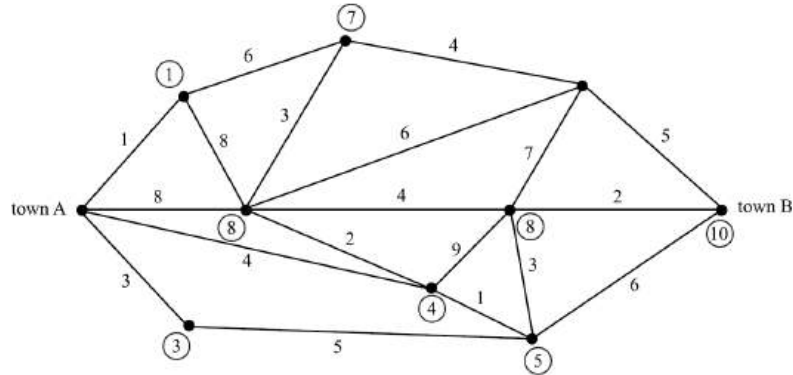
$\therefore$ Minimum spanning tree is:



$\Rightarrow D$

18. Networks, STD2 N2 2007 FUR1 4 MC

Using Dijkstra's algorithm:



Shortest path

$$= 4 + 1 + 3 + 2$$

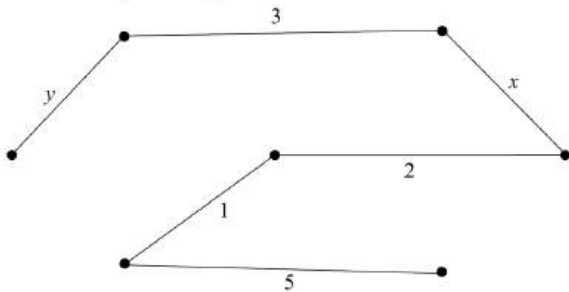
$$= 10$$

$\Rightarrow B$

19. Networks, STD2 N2 2007 FUR1 7 MC

Using Kruskal's Algorithm (as a guide),

Minimum spanning tree must be:



$$\therefore 19 = y + 3 + x + 2 + 1 + 5$$

$$8 = x + y$$

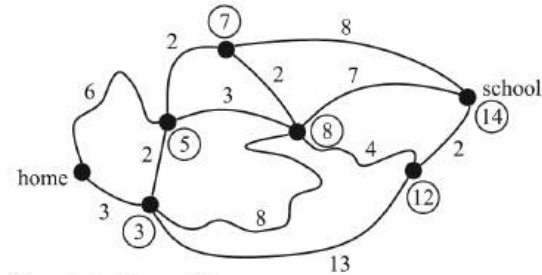
$\Rightarrow C$

20. Networks, STD2 N2 2011 FUR1 1 MC

$\Rightarrow B$

21. Networks, STD2 N2 2014 FUR1 3 MC

Using Dijkstra's algorithm:



Shortest time riding

$$= 3 + 2 + 3 + 4 + 2$$

$$= 14 \text{ minutes}$$

$\Rightarrow C$

22. Networks, STD2 N2 2020 FUR1 5 MC

One strategy - Using Prim's Algorithm

Starting at A

1st edge: $AH = 6$

2nd edge: $HG = 5$

then ... $AB = 7$, $GI = 9$, $IE = 6$, $EF = 5$

$DE = 8$, $CD = 7$

Total length = 53 m (not including x)

If $x < 9$, x would replace GI and minimum length would be less than 53m.

$\Rightarrow C$

23. Networks, STD2 N2 2021 HSC 2 MC

Working from A to E:

Sum of degrees = $4 + 3 + 4 + 2 + 3$

$$= 16 \quad \Rightarrow D$$

COMMENT: This simple question caused problems for many with mean mark just 58%.

24. Networks, STD2 N2 SM-Bank 25 MC

Possible paths:

$$W - Z - X - Y = 120 + 70 + 140 = 330 \text{ m}$$

$$W - Z - Y - X = 120 + 100 + 140 = 360 \text{ m}$$

$$W - X - Z - Y = 150 + 70 + 100 = 320 \text{ m}$$

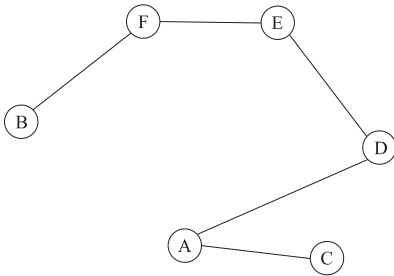
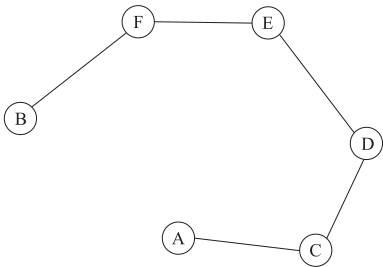
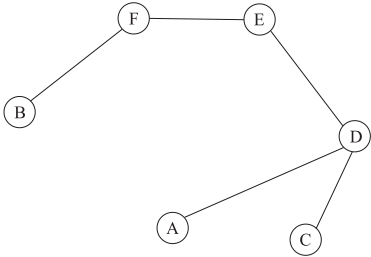
$$W - X - Y - Z = 150 + 140 + 100 = 390 \text{ m}$$

$\Rightarrow B$

25. Networks, STD2 N2 SM-Bank 7 MC

Since there are 6 vertices, each spanning tree will have 5 edges (with no cycles).

Consider if A and B are not connected, possible spanning trees are:



3 other spanning trees exist when BF is not connected, and likewise when EF and ED are not connected.

Finally, 2 other spanning trees exist when AD and AC are removed, and when AD and CD are removed.

$$\begin{aligned}\therefore \text{Total spanning trees} &= 4 \times 3 + 2 \\ &= 14\end{aligned}$$

26. Networks, STD2 N2 2016 FUR1 4 MC

Include k as a given edge before using Kruskal's Algorithm,

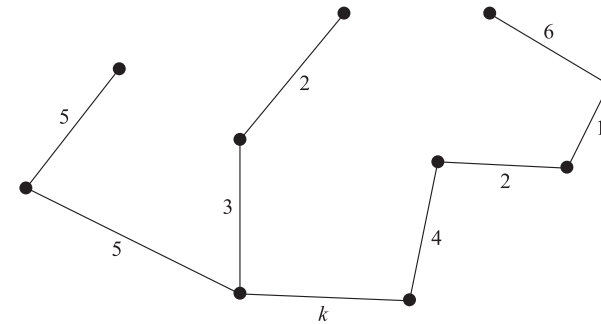
Edge 1: k

Edge 2: 1

Edges 3-4: 2

Edge 4: 3 etc...

$\therefore$ Minimum spanning tree



$$\text{Total weight} = k + 1 + 2 + 2 + 3 + 4 + 5 + 5 + 6$$

$$33 = k + 28$$

$$\therefore k = 5$$

$\Rightarrow D$

27. Networks, STD2 N2 2020 HSC 9 MC

Vertices = players

Edges = games between 2 players

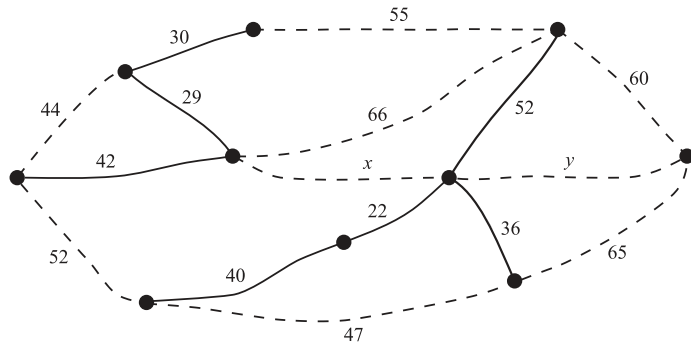
Since each player plays once against the three players in the other team, each vertex must be degree 3.

$\Rightarrow B$

♦ Mean mark 38%.

28. Networks, STD2 N2 2021 FUR1 8 MC

A partial minimal spanning tree can be drawn:



Consider each option:

A: If $x = 50$ (include), $y = 55$ (include)

→ Total length = $251 + 50 + 55 \neq 361$ (incorrect)

B: If $x = 50$ (include), $y = 60$ (include)

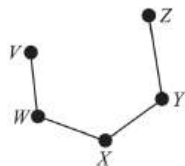
→ Total length = $251 + 50 + 55 = 356$ (correct)

Similarly, options C, D, E can be shown to be incorrect.

⇒ B

29. Networks, STD2 N2 2009 FUR1 8 MC

Consider an example of the graph described (below):



◆◆◆ Mean mark 25%.

A is possible – join V and Z

B is possible – join V and Z

C is possible – join W and Y

D is NOT possible

⇒ D

30. Networks, STD2 N2 2011 FUR2 1

a. Farnham to Carrie (shortest)

$$= 60 + 140$$

$$= 200 \text{ km}$$

b. Different paths are

FDC, FEDC, FEBC,

FEABC, FDEBC,

FDEABC

∴ 6 different ways

c. A possible path is *DFEABCDEB* and will finish

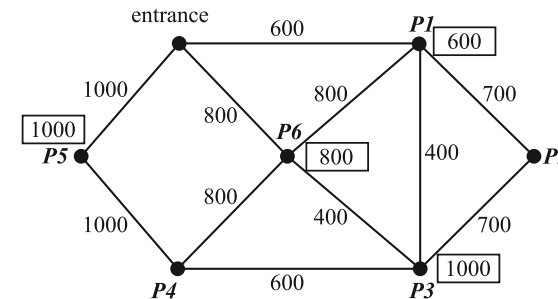
at Bredon - the only other odd-degree vertex.

(Note that solving this can be done quickly by applying the concept underlying the Königsberg Bridge problem.)

31. Networks, STD2 N2 2013 FUR2 1

a. 3

b. Using Dijkstra's algorithm:



Shortest distance

$$= E - P1 - P3$$

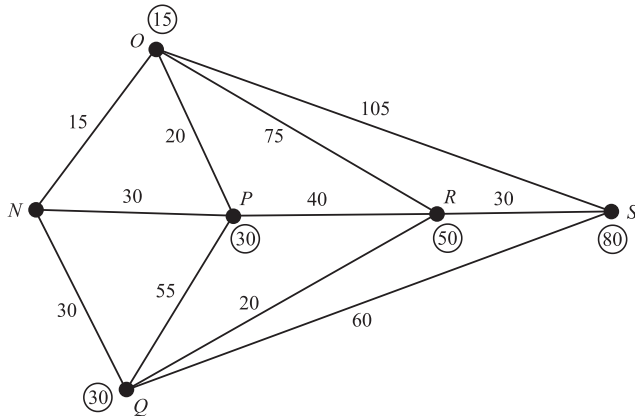
$$= 600 + 400$$

$$= 1000 \text{ m}$$

32. Networks, STD2 N2 2017 FUR2 1

a. Cost = 15 + 105
= \$120

b. Using Dijkstra's algorithm:



Fastest route is: *NQRS*.

∴ Other towns are Quigley and Rosebush.

33. Networks, STD2 N2 2011 FUR2 1

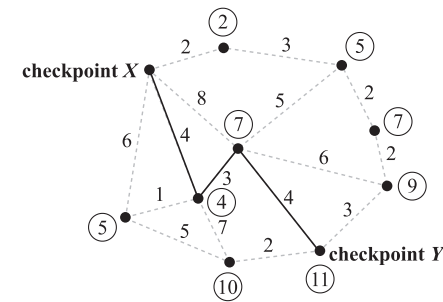
a. Farnham to Carrie (shortest)
= 60 + 140
= 200 km

b. Different paths are
FDC, FEDC, FEBC,
FEABC, FDEBC,
FDEABC

∴ 6 different ways

34. Networks, STD2 N2 2010 FUR2 2

Using Dijkstra's Algorithm:



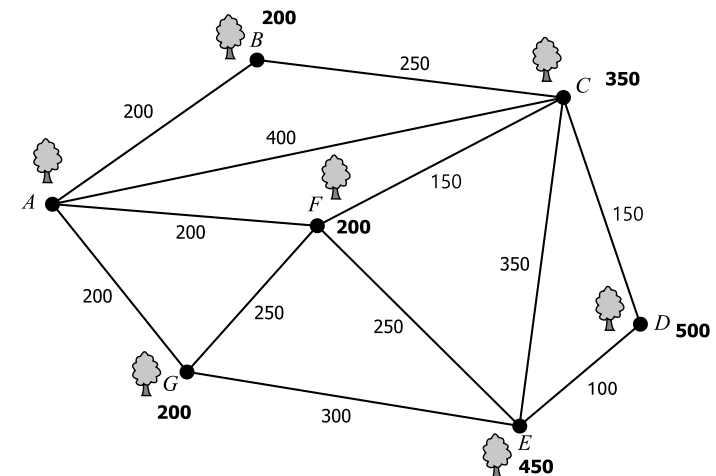
Shortest distance *X* to *Y*

$$= 4 + 3 + 4$$

$$= 11 \text{ minutes}$$

35. Networks, STD2 N2 SM-Bank 13

One strategy - using Dijkstra's algorithm:



Shortest path is *AFCD*

$$\text{Shortest distance} = 200 + 150 + 150$$

$$= 500 \text{ metres}$$

36. Networks, STD2 N2 SM-Bank 20

	A	B	C	D
A	0	4	2	-
B	4	0	2	6
C	2	2	0	3
D	-	6	3	0

37. Networks, STD2 N2 2015 FUR2 1

a. \$300

b. Minimum cost of K to N

$$= 440 + 480$$

$$= \$920$$

c. N and P (or P and N)

COMMENT: Part (c) requires an understanding of the Königsberg bridge problem!

(The inspector must start from and finish on vertices of odd degree)

d.i. Using Prim's Algorithm:

Starting at Vertex L

1st Edge: $L \rightarrow M$ (300)

2nd Edge: $L \rightarrow K$ (360)

3rd Edge: $K \rightarrow J$ (250)

4th Edge: $J \rightarrow P$ (200) etc...

d.ii. Disconnect $J - P$ and $O - P$

$$\text{Savings} = 200 + 400 = \$600$$

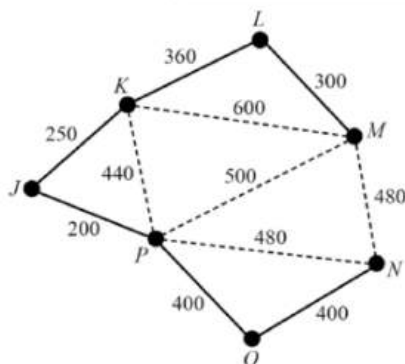
Add in $M - N$

$$\text{Cost} = \$480$$

$$= 600 - 480$$

$\therefore$ Net savings

$$= \$120$$



MARKER'S COMMENT: Many students had difficulty finding the minimum spanning tree, often incorrectly excluding PO or KL .

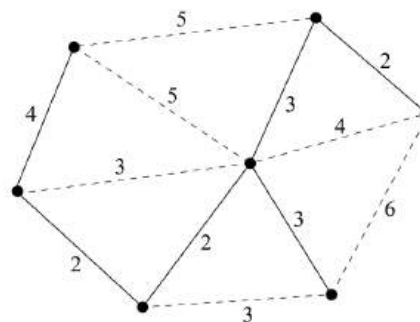
38. Networks, STD2 N2 2008 FUR2 1

a. Using Kruskal's Algorithm

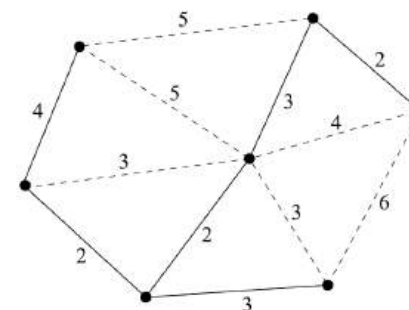
Edges 1-3: 2

Edges 4-5: 3 (2 edges with weight 3 create a circuit and are ignored)

Edge 6: 4



OR



b. Length of minimal spanning tree

$$= 2 + 2 + 2 + 3 + 3 + 4$$

$$= 16 \text{ metres}$$

c. 2

39. Networks, STD2 N2 2011 FUR2 2

a. Using Prim's Algorithm

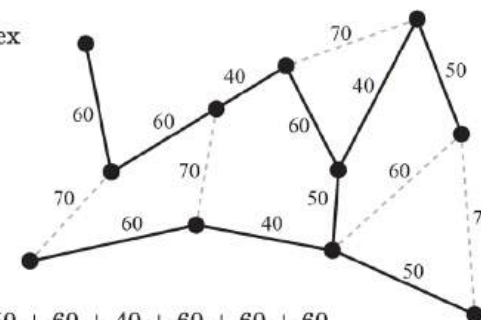
Starting at bottom right vertex

1st Edge: 50

2nd Edge: 40

3rd Edge: 50

4th Edge: 40 etc...

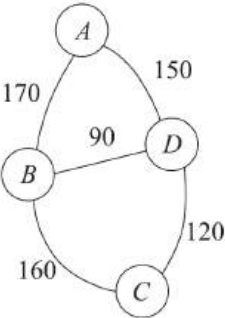


b. Total length of water pipe

$$= 50 + 40 + 50 + 40 + 50 + 60 + 40 + 60 + 60 + 60$$

$$= 510 \text{ metres}$$

40. Networks, STD2 N2 SM-Bank 28

- a.
- 
- b. Shortest Path from A (visiting all stations)
 $A - B - D - C$
 Distance = $170 + 90 + 120$
 $= 380 \text{ km}$

41. Networks, STD2 N2 SM-Bank 2

- i. One Strategy: Using Prim's Algorithm

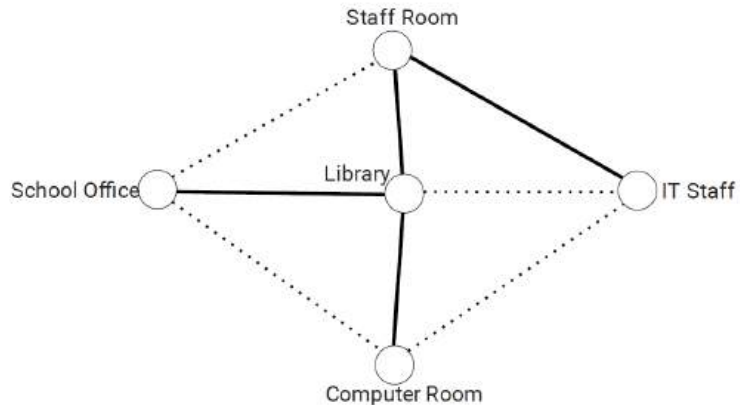
Starting vertex - Staff Room

1st edge: Staff Room - Library

2nd edge: Library - School Office

3rd edge: Staff Room - IT Staff

4th edge: Library - Computer Room



- ii. Minimum cost = $300 + 300 + 400 + 500$
 $= \$1500$

42. Networks, STD2 N2 SM-Bank 14

- i. 2 (the house and the top right vertex)

- ii. One Strategy - Using Prim's algorithm:

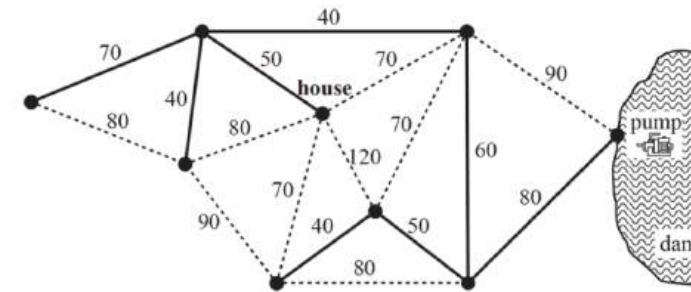
Starting at the house

1st edge: 50

2nd edge: 40 (either)

3rd edge: 40

4th edge: 60 etc...



- iii. Minimum spanning tree

43. Networks, STD2 N2 SM-Bank 37

- i. ACT has 1 border (with NSW)

$\therefore$ Degree of ACT's vertex = 1

- ii. NSW is Vertex B (it is connected to the ACT - Vertex D)

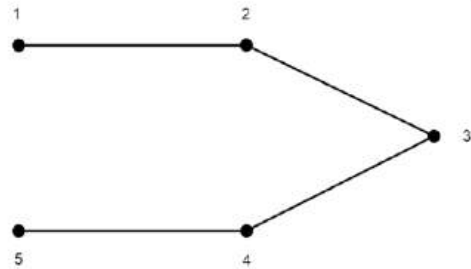
$\therefore$ Queensland is vertex A as it is connected to B and has degree 3.

(C is Victoria as it has degree 2)

MARKER'S COMMENT: Many students, surprisingly, had problems with part (i).

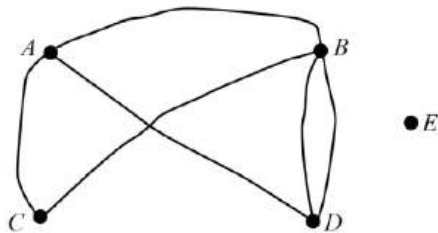
44. Networks, STD2 N2 2007 FUR2 1

- a. Minimum number of edges = 4
- b. One of many possibilities,



45. Networks, STD2 N2 2009 FUR2 1

- a. E has no land borders with other suburbs.
- b.i. & ii.



46. Networks, STD2 N2 2017 FUR2 3

- a.i. Minimum spanning tree.
- a.ii Using Kruskal's Algorithm

Edge 1: 50

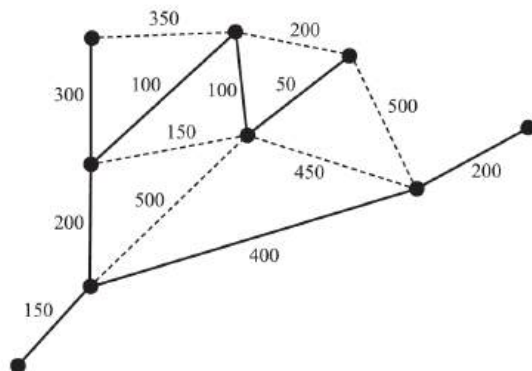
Edges 2-3: 100

Edge 4: 150 (ignore the 150 edge that creates a circuit)

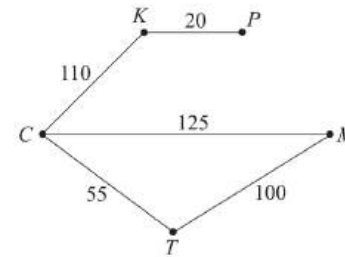
Edges 5-6: 200 (ignore the 200 edge that creates a circuit)

Edge 7: 300

Edge 8: 400



47. Networks, STD1 N1 2019 HSC 17



Edge weights are in minutes duration.

48. Networks, STD2 N2 2019 HSC 30

- a. One strategy - using Prim's algorithm:

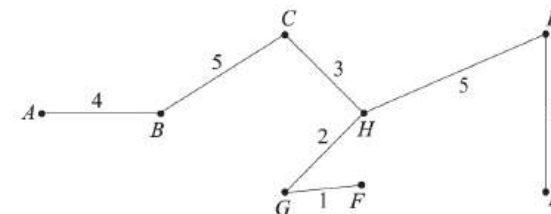
Starting at A

1st edge - AB , 2nd edge - BC

3rd edge - CH , 4th edge - HG

5th edge - GF , 6th edge - HD

7th edge - DE or HE

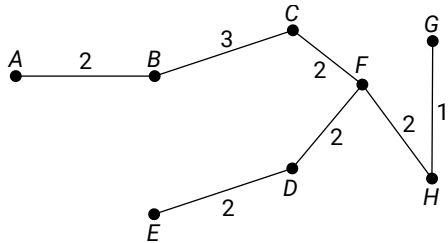


Maximum length = $4 + 5 + 3 + 2 + 1 + 5 + 5 = 25$ km

- b. Shortest Path is $CGHE$

a. Using Kruskal's Algorithm (one of many possibilities):

Edge 1 : GH (1)
 Edge 2 : FH (2)
 Edge 3 : CF (2)
 Edge 4 : FD (2)
 Edge 5 : DE (2)
 Edge 6 : BC (3)
 Edge 7 : AB (2)



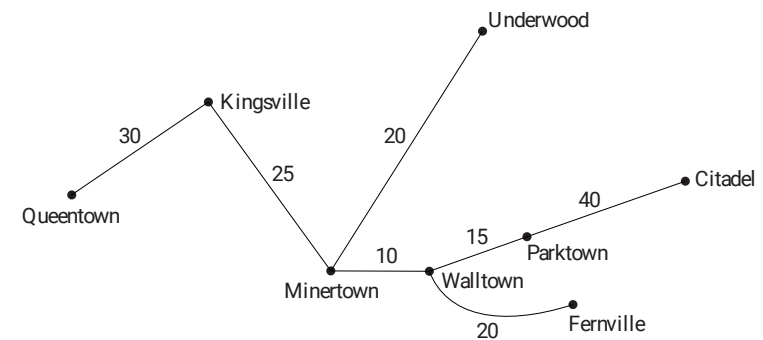
$$\begin{aligned}\text{Minimum length of spanning tree} &= 1 + 2 + 2 + 2 + 2 + 3 + 2 \\ &= 14\end{aligned}$$

b. Add CK to the minimum spanning tree in (a).

$$\begin{aligned}\therefore \text{Revised length} &= 14 + 10 \\ &= 24\end{aligned}$$

a. Using Prim's algorithm (starting at W):

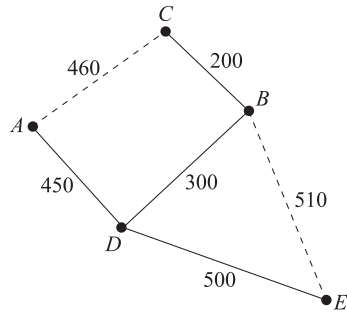
1st edge: WM
 2nd edge: WP
 3rd/4th edges: MU and WF
 5th edge: MK
 6th edge: KQ
 7th edge: PC



$$\text{Length of minimum spanning tree} = 160$$

$$\begin{aligned}\text{b. Fastest route } (Q \text{ to } U) &= 45 + 20 \\ &= 65 \text{ minutes}\end{aligned}$$

51. Networks, STD2 N2 SM-Bank 30



Consider Kruskal's algorithm for the given minimum spanning tree:

$CB \rightarrow$ 1st edge (200 - minimum weight)

$BD \rightarrow$ 2nd edge (300)

$AD \rightarrow$ 3rd edge (450)

$\Rightarrow AC = 460$ (choose any value such that $AC > 450$)

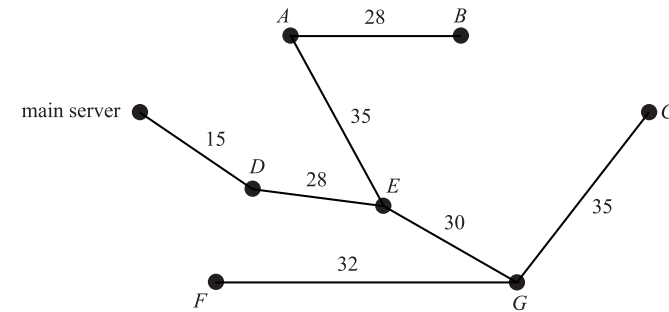
$DE \rightarrow$ 4th edge (500)

$\Rightarrow BE = 510$ (choose any value such that $BE > 500$)

$\therefore BE > 500$ (BE must be greater than DE)

$\therefore AC > 450$ (AC must be greater than AD)

52. Networks, STD2 N2 SM-Bank 34



Using Prim's Algorithm:

Edge 1: main server to D (15)

Edge 2: DE (28)

Edge 3: EG (30)

Edge 4: GF (32)

Edge 5: EA (35)

Edge 6: AB (28)

Edge 7: GC (35)

$\therefore$ Minimum length

$$= 15 + 28 + 30 + 32 + 35 + 28 + 35$$

$$= 203 \text{ m}$$

53. Networks, STD2 N2 SM-Bank 3

One Strategy - Kruskal's algorithm

There are 6 vertices, so we need 5 edges.

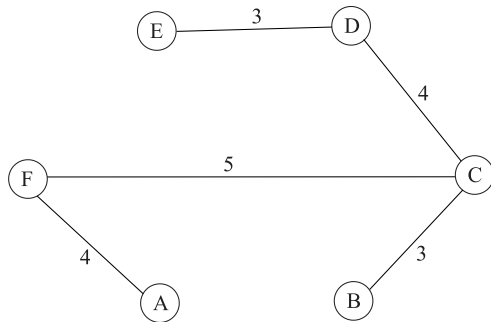
Edge 1: BC (least weight) - could have chosen ED

Edge 2: DE (next least weight)

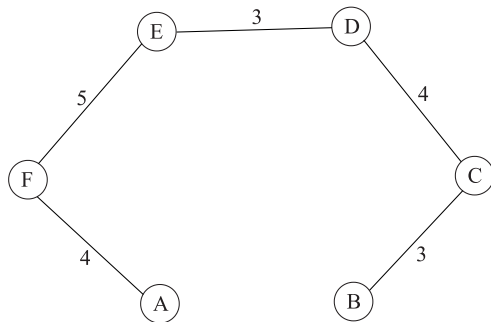
Edge 3: AF (could have been CD)

Edge 4: CD

Edge 5: CF or EF (reject CE as it creates a cycle)



or



54. Networks, STD2 N2 SM-Bank 4

One Strategy: Using Prim's algorithm

6 vertices → need 5 edges.

Starting at vertex A (any vertex can be chosen)

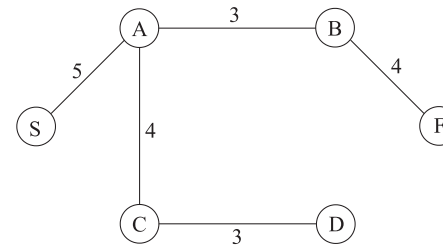
1st Edge: AB

2nd Edge: AC (BF also possible)

3rd Edge: CD

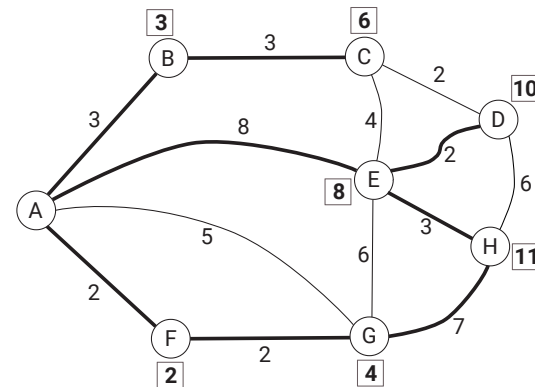
4th Edge: BF

5th Edge: AS



55. Networks, STD2 N2 SM-Bank 5

One Strategy: Dijkstra's algorithm



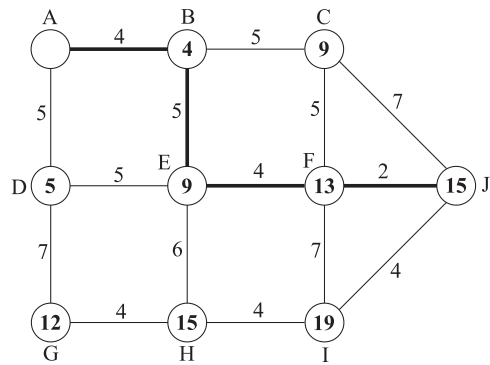
Shortest distance paths:

A - E - H or A - F - G - H

Shortest distance is 11 km.

56. Networks, STD2 N2 SM-Bank 6

One Strategy: Using Dijkstra's algorithm



Shortest path is:

A - B - E - F - J

∴ Weight of shortest path

$$= 4 + 5 + 4 + 2$$

$$= 15$$

57. Networks, STD2 N2 SM-Bank 8

One Strategy - Using Prim's algorithm:

6 vertices → 5 edges on spanning tree

Starting at vertex A

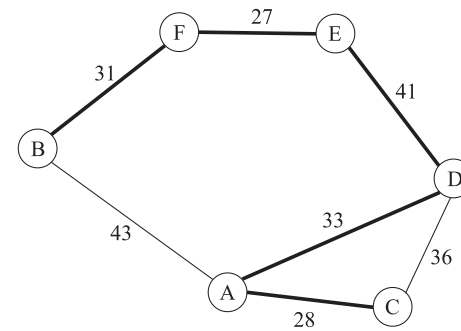
Edge 1: AC

Edge 2: AD

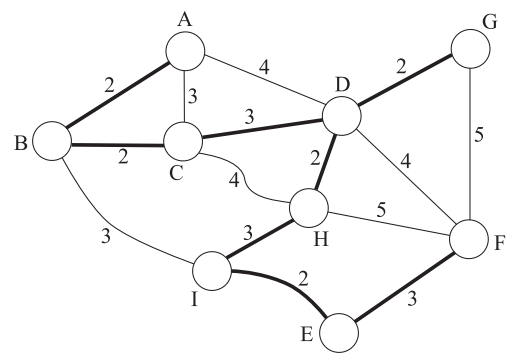
Edge 3: ED (reject CD which creates a cycle)

Edge 4: EF

Edge 5: BF



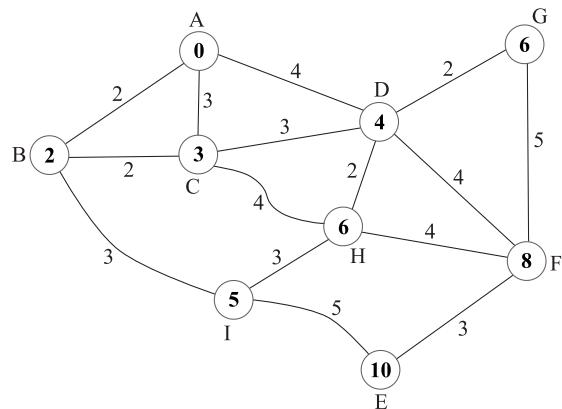
58. Networks, STD2 N2 SM-Bank 9



One strategy - Using Kruskal's algorithm:
Edges 1 - 5: AB, BC, DG, DH and EI
Edges 6 - 8: CD, EF and HI
(note AC cannot be chosen → creates a cycle)
NB: There is more than one minimal spanning tree in this circuit (having a weight of 19).

59. Networks, STD2 N2 SM-Bank 10

One strategy - Using Dijkstra's algorithm:



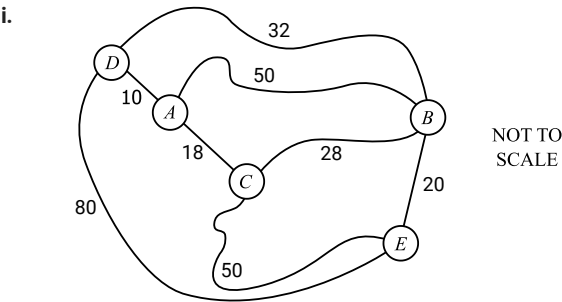
The shortest path A - B - I - E has a distance (weight) of 10.

60. Networks, STD2 N2 SM-Bank 11

	A	B	C	D	E
A	0	10	17	10	10
B	10	0	25	—	12
C	17	25	0	—	—
D	10	—	—	0	17
E	10	12	—	17	0

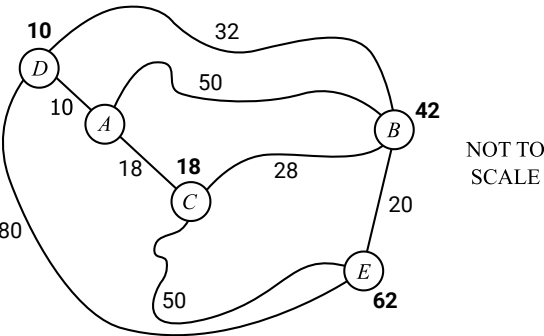
Note the symmetry in this table across the diagonal.

61. Networks, STD2 N2 SM-Bank 12



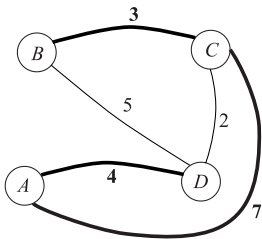
Important to note that network diagrams do not need to be drawn to scale.

ii. One strategy - using Dijkstra's algorithm:

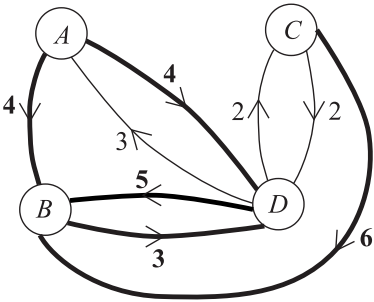


∴ Shortest travelling time is the path $A - D - B - E$
= 10 + 32 + 20
= 62 minutes

62. Networks, STD2 N2 SM-Bank 21



63. Networks, STD2 N2 SM-Bank 22



64. Networks, STD2 N2 SM-Bank 23

		To			
		A	B	C	D
From	A	-	3	0	0
	B	2	-	0	5
	C	0	4	-	4
	D	0	7	0	-